Computer Networks

MCA5122

Syllabus

MCA 5122 COMPUTER NETWORKS [4 0 0 4]

Introduction to Computer Networks: Definition, Network Layer, Network Layer services, Interfacing - Bridges, IP addressing, Subnetting, IPv6 addressing, Delivery Forwarding, and Routing of IP Packets, Internet Protocol - Datagram, Fragmentation, Options, Checksum, Introduction to Routing Protocols, Interior and Exterior routing, Dynamic IP Routing Protocols - RIP, OSPF, Routing between peers – BGP, ARP, Internet Control Message Protocol, User Datagram Protocol, Transmission Control Protocol and Introduction to application layer, Domain Name System (DNS), and DHCP

SDL: Introduction to application layer, Domain Name System (DNS), and DHCP.

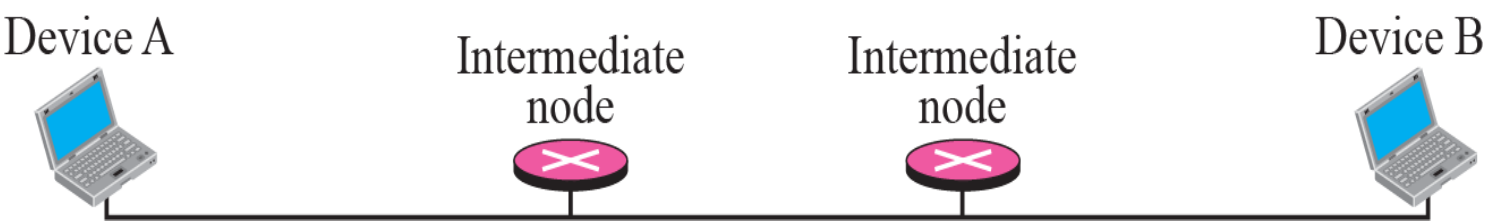
References:

1. Forouzan B. A., *TCP/IP Protocol Suite (4e),* Tata McGraw Hill 2017.
2. Tanenbaum A. S., *Computer Network (5e),* Prentice Hall of India Pvt Ltd 2013.
3. Forouzan B. A., *Data Communications and Networking (6e),* Tata McGraw Hill 2022.
4. Garcia L., Widjala, *Communication Networks (2e),* Tata McGraw Hill 2004.

The problem

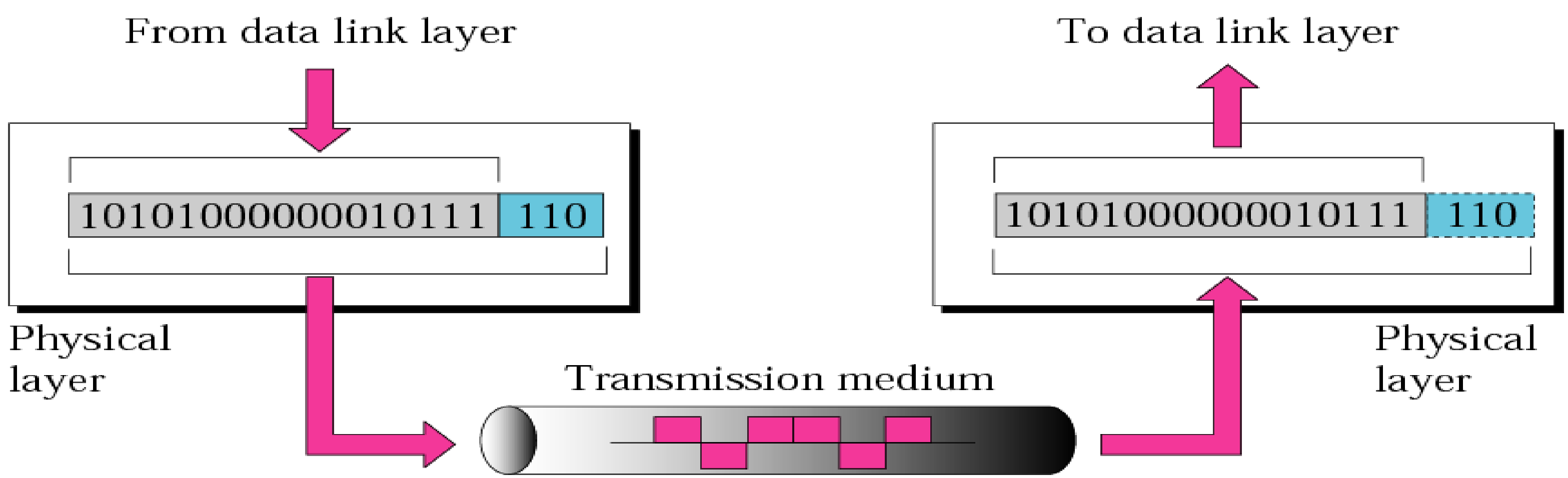
- Let us consider the problem of two computers hooked together & talk to each other.

A big file we want to send from A to B



1st Issue

- Physical Media (phone lines, cables)
- how to transmit bits ?
  - Bits to Analog signals
    - Signal encoding/ decoding techniques
      - Data rate, Duration or length of a bit

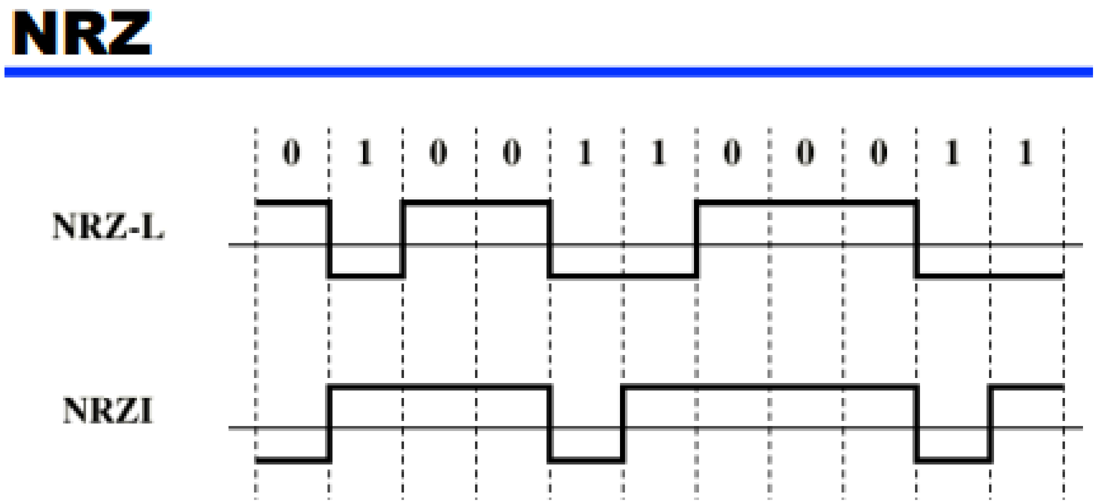


These functionalities are categorized as – Physical Layer functionalities

Signal encoding/ decoding

Nonreturn to Zero-Level (NRZ-L

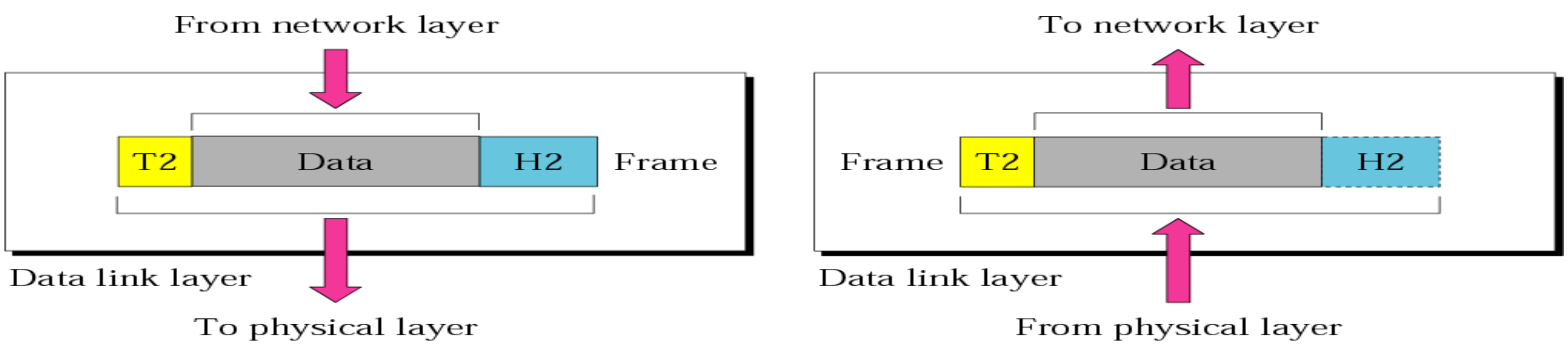
Nonreturn to Zero Inverted



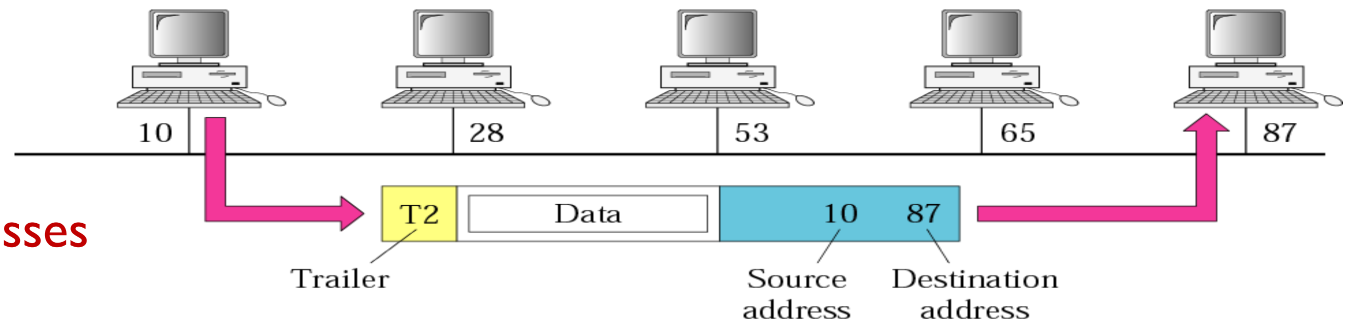
2nd Issue-

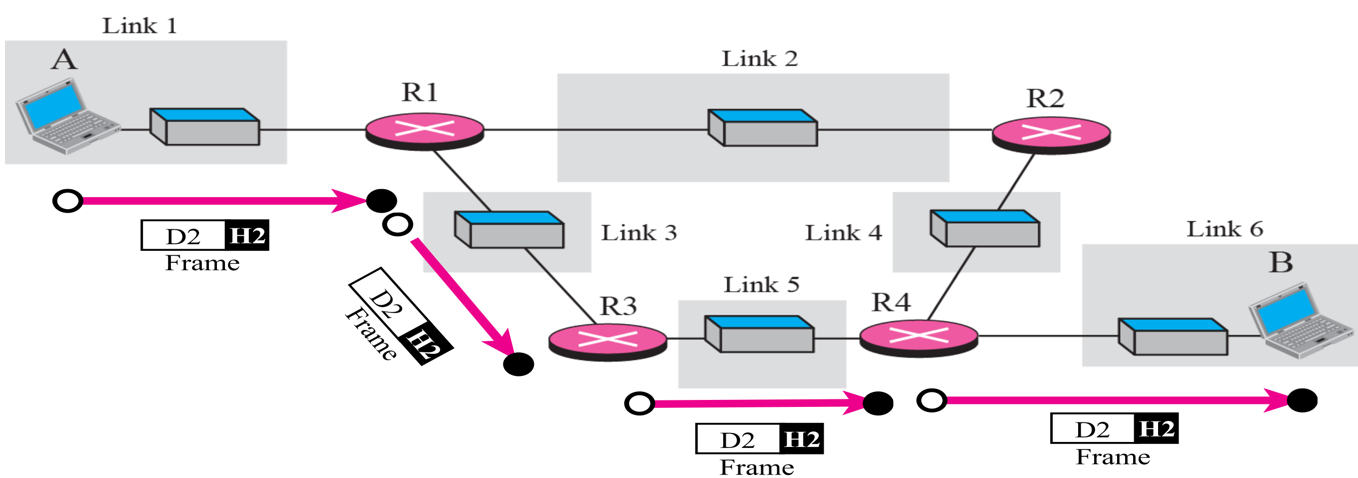
- Framing
  - Sending entire file as bit stream is not efficient.
  - One method is –
    - Make packs of smaller size-'frames' and ship.
  - Physical media is not Nosie-free
    - Error detection and corrections.
- Sharing media
    - CSMA/CD, ALOHA , Token passing protocols
  - Physical address is used to deliver to immediate next device (Hop-to-Hop Delivery)

Immediate next device may be destination Computer or any other intermediate device



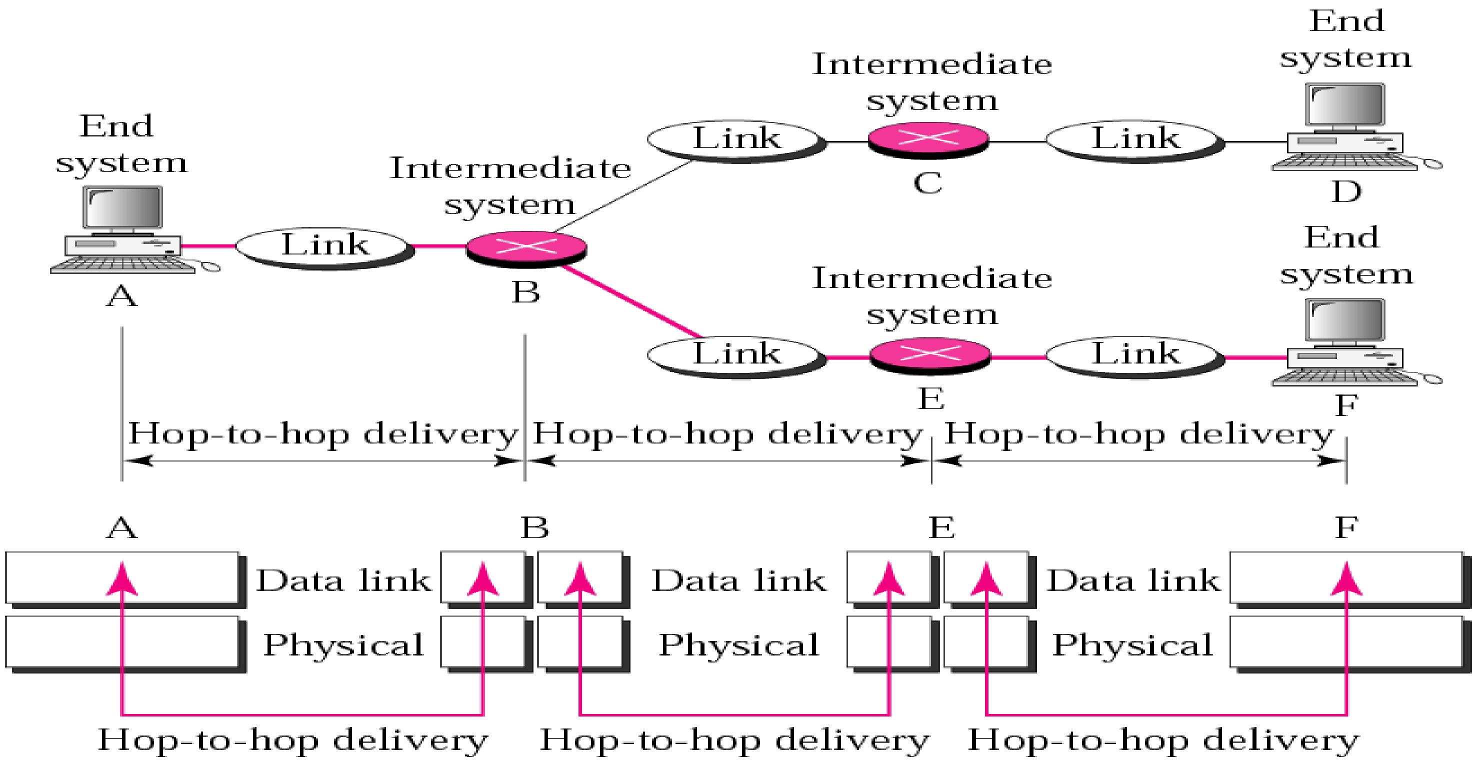
Physical Addresses





These functionalities are categorized as – Data Link Layer functionalities

Now we need 2 layers functionalities



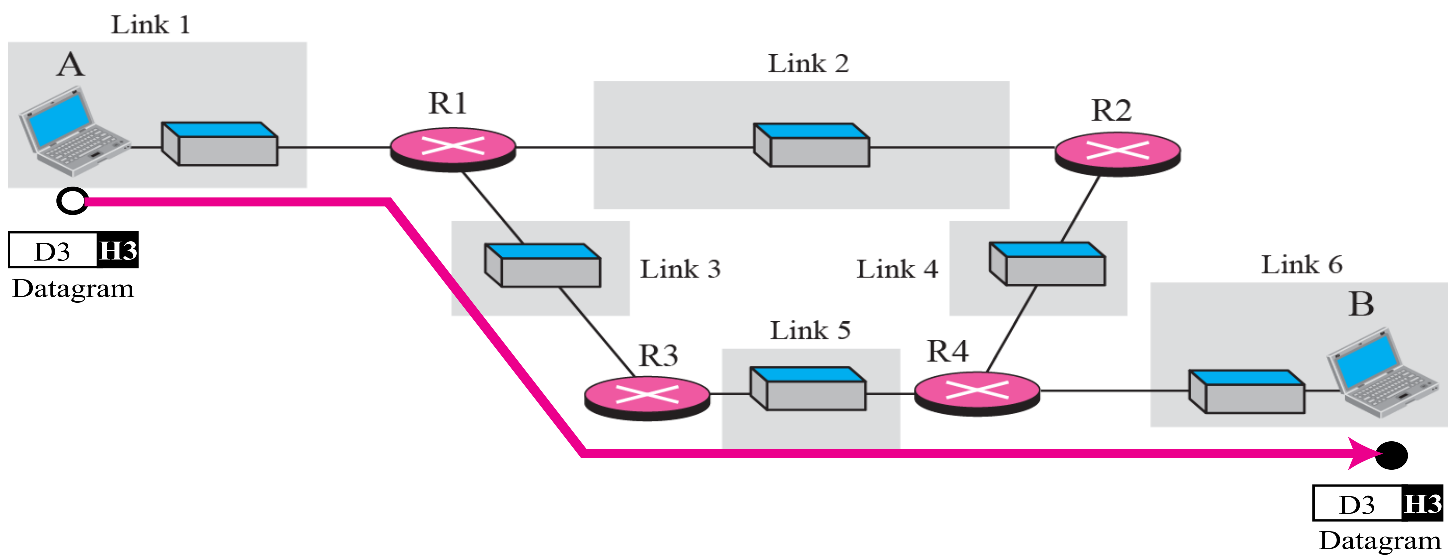
Syllabus:

Media Access sub layer(MAC) and LANS:

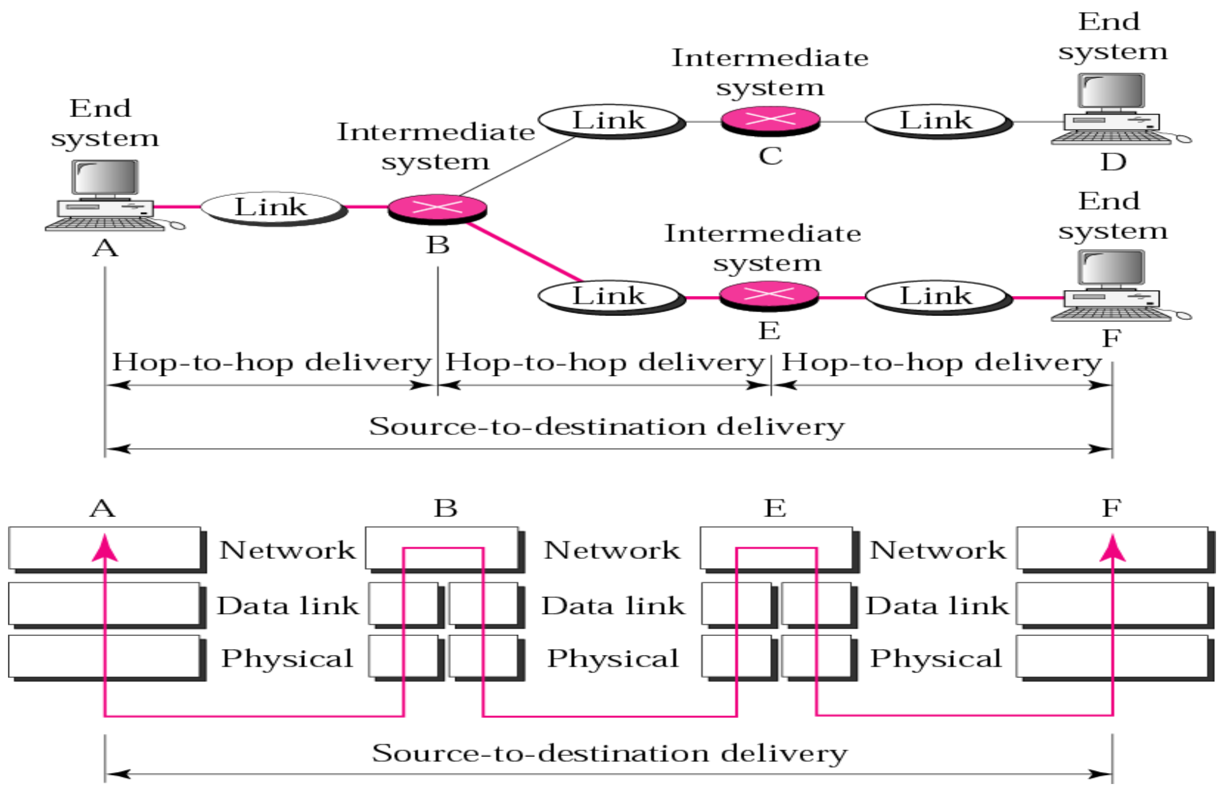
IEEE 802.3 frame format. Approaches to sharing transmission Medium, Random Access Protocols, CSMA/CD (Token Passing protocols), IEEE LAN standards, Bridges, Switches and Routers.

3rd Issue

- How to reach destination Computer ?
  - Because, in reality, there may be many intermediate devices & link technologies between Sender and Receive.
    - Support for Heterogeneous Link Technology -MTU, fragmentation
    - Need for Logical Address. –IPv4, IPv6
    - Routing –OSPF, RIP, BGP protocols



- Extra Functionalities needed now are-
  - Logical Addressing & Routing



These functionalities are categorized as – Network Layer functionalities

Now we need 3 layers functionalities- Network Layer

Source to Destination –means Communication from Computer A to D is achieved ; B ,C E are intermediate devices(Routers)

Syllabus:

Network Layer:
Internal Organization of NL.

IP addressing:
Decimal Notation, Classes, Special Addresses, Unicast multicast and broadcast addresses, applying for IP address, Private networks.

Subnetting and Supernetting:
Subnetting, Masking, Variable length subnetting, supernetting.

Delivery Forwarding, and Routing of IP Packets:
Connection – oriented v/s connectionless services. Direct Vs. Indirect Delivery, Forwarding, Routing methods, Static Vs. Dynamic routing, Routing module and Routing table design.

Syllabus:

Internet Protocol (IP): Datagram, Fragmentation, Options, Checksum & IP Design.

ARP and RARP: ARP, ARP design & RARP.

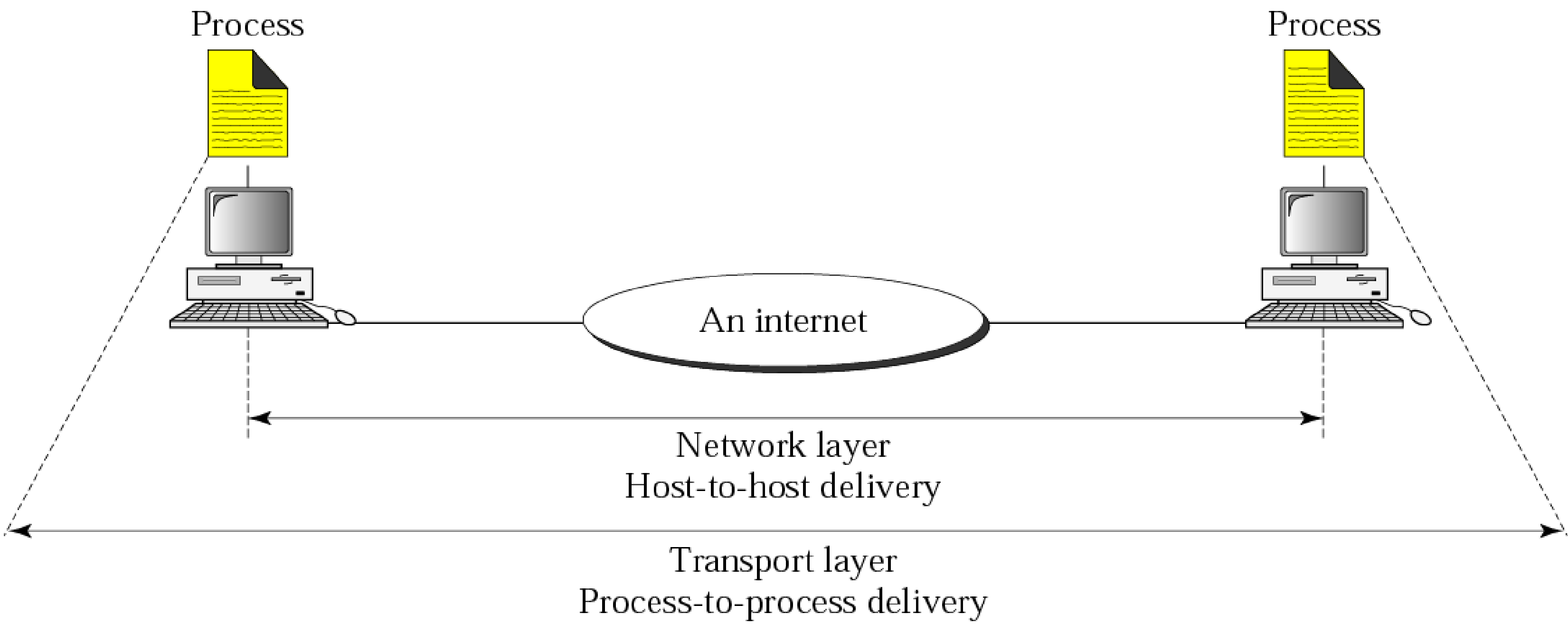
Internet Control Message Protocol (ICMP):
Types of messages, message format, error reporting, query, Checksum& ICMP Design.

Internet Group Management Protocol (IGMP):
Multicasting, IGMP, Encapsulation, Multicast backbone & IGMP design.

Introduction to Routing Protocols:
Interior and Exterior routing, RIP, RIP Version 2, OSPF & BGP.

4^{th} Issue

- How to Deliver to destination Process ?



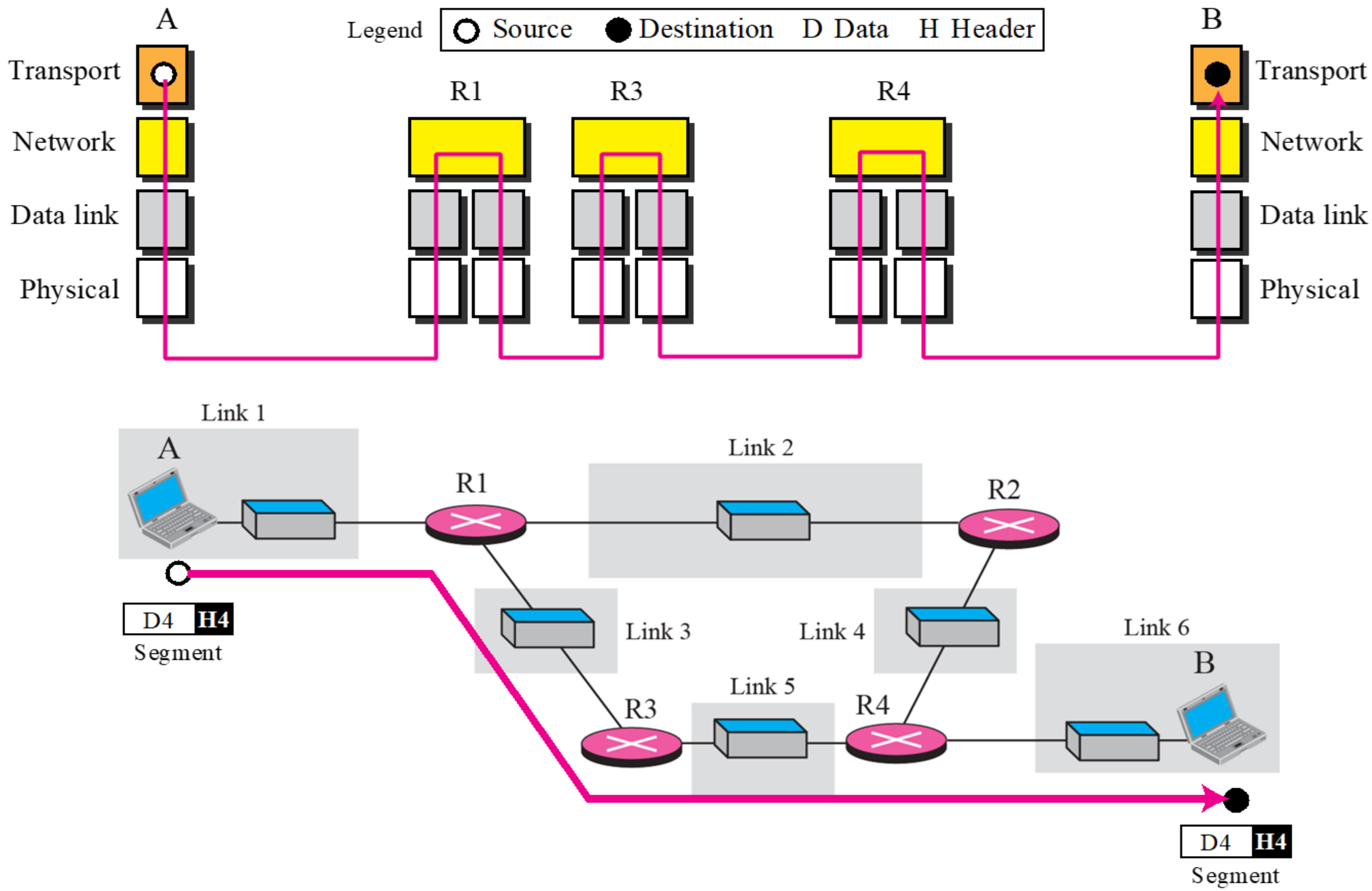
Main functionalities Required are-

Process-to-Process delivery

Flow Control

Error Control

Reassembly



Now we need 4 layers functionalities

Syllabus:

User Datagram Protocol (UDP) :

Process-To-Process Communication, User datagram, UDP operation, Uses of UDP.

Transmission Control Protocol (TCP) :

TCP services, A TCP connection, Flow control, Error Control, Congestion control, TCP Timer.

5^{th} Issue

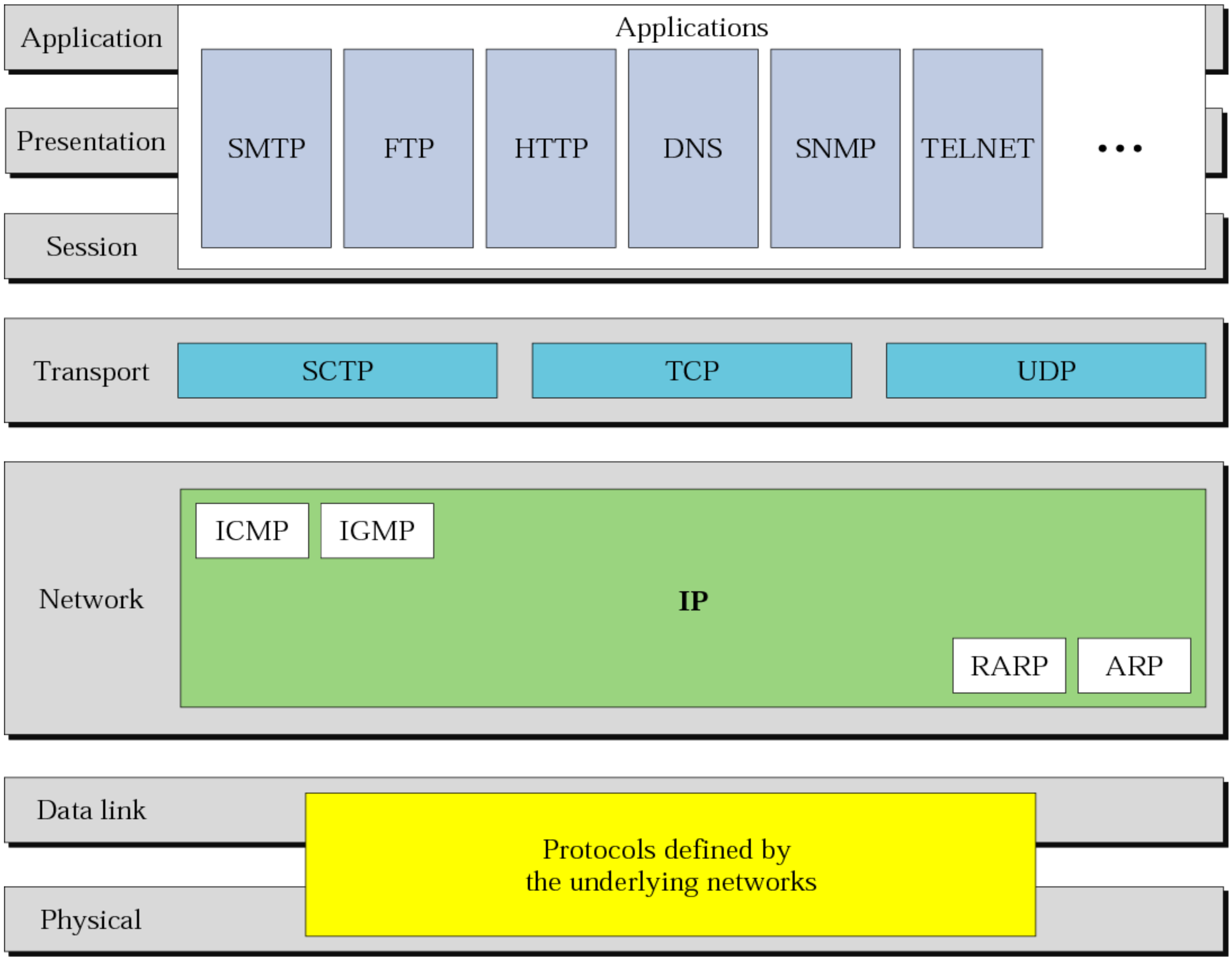
The session layer is the network dialog controller. It establishes, maintains, and synchronizes the interaction between communicating systems.

6^{th} Issue

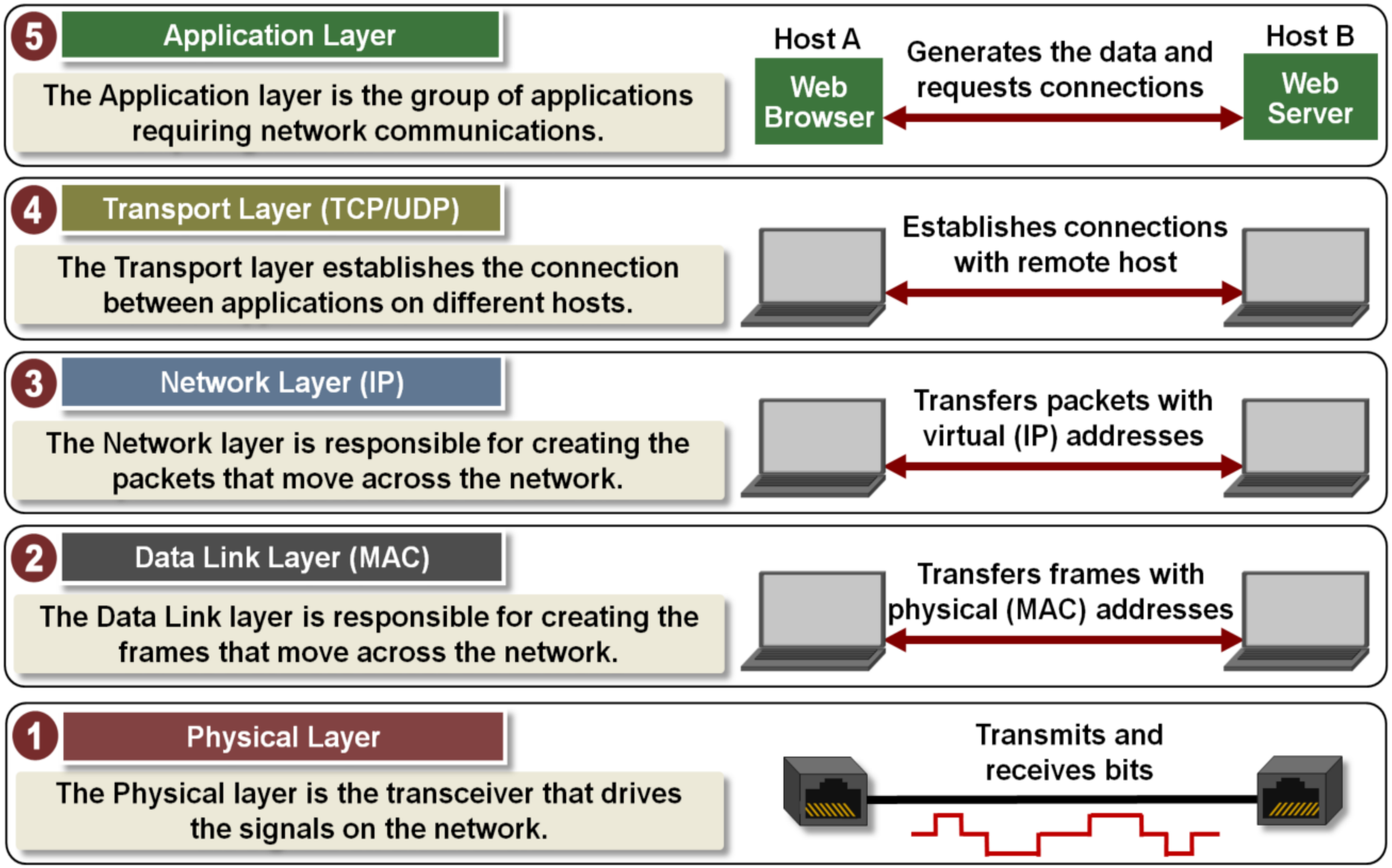
The presentation layer is concerned with the Translation, Encryption/Decryption , Compression/Decompression of the information exchanged between two systems.

7th Issue

- The application layer provides the interfaces and services to access the network

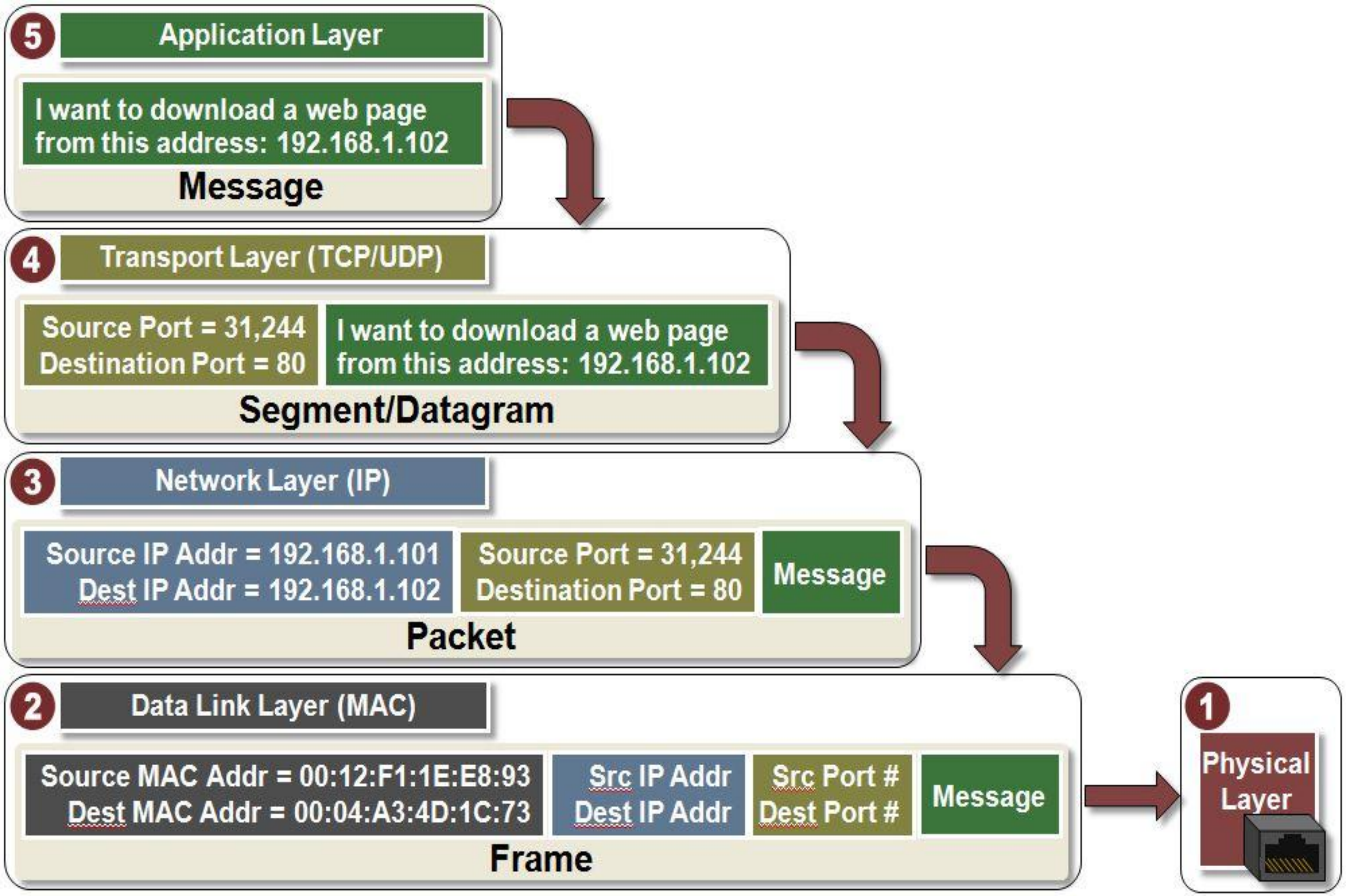

Application
Presentation
Session
Applications
SMTP
FTP
HTTP
DNS
SNMP
TELNET
• • •
Transport
SCTP
TCP
UDP
Network
ICMP
IGMP
IP
RARP
ARP
Data link
Physical
Protocols defined by
the underlying networks

Summary



5
Application Layer
The Application layer is the group of applications requiring network communications.
Host A
Web Browser
Generates the data and requests connections
Host B
Web Server
4
Transport Layer (TCP/UDP)
The Transport layer establishes the connection between applications on different hosts.
Establishes connections with remote host
3
Network Layer (IP)
The Network layer is responsible for creating the packets that move across the network.
Transfers packets with virtual (IP) addresses
2
Data Link Layer (MAC)
The Data Link layer is responsible for creating the frames that move across the network.
Transfers frames with physical (MAC) addresses
1
Physical Layer
The Physical layer is the transceiver that drives the signals on the network.
Transmits and receives bits

Summary


5
Application Layer
I want to download a web page from this address: 192.168.1.102
Message
4
Transport Layer (TCP/UDP)
Source Port = 31,244
Destination Port = 80
I want to download a web page from this address: 192.168.1.102
Segment/Datagram
3
Network Layer (IP)
Source IP Addr = 192.168.1.101
Dest IP Addr = 192.168.1.102
Source Port = 31,244
Destination Port = 80
Message
Packet
2
Data Link Layer (MAC)
Source MAC Addr = 00:12:F1:1E:E8:93
Dest MAC Addr = 00:04:A3:4D:1C:73
Src IP Addr
Dest IP Addr
Src Port #
Dest Port #
Message
Frame
1
Physical Layer

Summary

Layer #	Layer Name	Protocol	Protocol Data Unit	Addressing
5	Application	HTTP, SMTP, etc…	Messages	n/a
4	Transport	TCP/UDP	Segments/ Datagrams	Port #s
3	Network or Internet	IP	Packets	IP Address
2	Data Link	Ethernet, Wi-Fi	Frames	MAC Address
1	Physical	10 Base T, 802.11	Bits	n/a

The upper layers are almost always implemented in software;

lower layers are a combination of hardware and software, except for the physical layer,

which is mostly hardware.

Text Books

- Behroua A. Forouzan – “TCP/IP PROTOCOL SUITE”, Tata McGraw Hill, Third Edition, 2010
- Behroua A. Forouzan – “Data Communication and Networking”, Tata McGraw Hill, Fourth Edition, 2007
- Tannenbaum, A.S. – “COMPUTER NETWORKS”, Prentice Hall of India [EE Edition], 4th edition, 2003.
- Alberto Leon- Garcia – “Communication Networks”, Tata McGraw Hill, 2nd edition, 2004